

	$F = 2.4860 \approx 2.5 \text{ N}$
2(a)	At maximum height, kinetic energy at the top will be minimum as the vertical component of velocity will be zero. $KE_{Top} = \frac{1}{2}m(u_x)^2$ $65 = \frac{1}{2}(0.55)(u_x)^2$ $u_x = 15.374 \text{ m s}^{-1}$ $\tan(42^\circ) = \frac{u_y}{u_x}$ $\tan(42^\circ) = \frac{v_y}{15.374}$ $v_y = 13.843 \approx 14 \text{ m s}^{-1}$
2(b)	Using $s = ut + \frac{1}{2}at^2$ $0 = 13.843t + \frac{1}{2}(-9.81)t^2$ $t = 2.8222 \text{ s}$ Horizontal displacement = $u_x t = 15.374(2.8222) = 43.388 \approx 44 \text{ m}$
2(c)	Time to reach maximum height is smaller than time taken to return to ground from maximum height. Object moving up has <u>resistance acting in the same direction as weight</u>, hence a larger downward resultant force which result a <u>larger deceleration</u>. Object moving down has <u>resistance opposing weight</u>, hence a smaller resultant downward force which results a <u>smaller acceleration</u>. <u>Average speed moving upwards is therefore higher than average speed moving downwards.</u> Since upwards and downwards distance is the same, <u>the time to</u>

	<u>reach maximum height is smaller</u> than time taken to return to ground from maximum height.
2(c)(i)	For Geostationary Satellite, period of orbit = 24 hrs = 86400 s